SWE 261P Software Testing and Analysis - Part 3 Report

PDFsam Basic: White Box Testing and Coverage

JAVA 21

JUNIT 5

MAVEN

JACOCO 0.8.12

Course SWE 261P
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This report documents the **structural (white-box) testing** process of **PDFsam Basic**, covering JaCoCo code coverage configuration, baseline measurement, and targeted test improvements across three non-GUI modules.

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🎯 1. Structural Testing: Definition and Importance

1.1 What Is Structural Testing?

Structural testing (white-box testing) is a software testing technique that uses knowledge of a program's internal structure—its source code, control flow, and data flow—to design test cases. Unlike black-box testing, which treats the system as an opaque entity, structural testing examines the *implementation* to systematically exercise different code paths, branches, and conditions.

1.2 Why Structural Testing Matters

Benefit	Description
Reveals Hidden Defects	Tests derived from implementation details uncover bugs in error-handling paths, boundary conditions, and rarely-executed branches
Coverage Measurement	Tools like JaCoCo quantify how much code is exercised (line, branch, method coverage)
Complementary	Fills gaps left by specification-based (black-box) tests
Regression Safety	High structural coverage provides confidence that future changes don't introduce regressions

1.3 Coverage Metrics

Common structural coverage criteria include:

- **Line Coverage**: Percentage of executable statements executed
- **Branch Coverage**: Percentage of decision outcomes (true/false) evaluated
- **Method Coverage**: Percentage of methods invoked
- **Instruction Coverage**: Percentage of bytecode instructions executed

1.4 Tool: JaCoCo

In order to see the test coverage of the current tests in the application, we used a tool called Java Code Coverage (JaCoCo). This plugin can be installed by including it in the pom.xml file.

```xml
<plugin>
    <groupId>org.jacoco</groupId>
    <artifactId>jacoco-maven-plugin</artifactId>
    <version>0.8.12</version>
    <executions>
        <execution>
            <id>prepare-agent</id>
            <goals><goal>prepare-agent</goal></goals>
        </execution>
        <execution>
            <id>report</id>
            <phase>test</phase>
            <goals><goal>report</goal></goals>
            <configuration>
                <formats>
                    <format>CSV</format>
                    <format>XML</format>
                </formats>
            </configuration>
        </execution>
    </executions>
</plugin>
```

📊 2. Existing Test Suite Coverage — Baseline

We ran `mvn clean test` with JaCoCo on three non-GUI modules **before** adding any new tests to establish a baseline.

2.1 Running Baseline Coverage

```
# Run all tests with JaCoCo
mvn clean test jacoco:report -pl pdfsam-model,pdfsam-core,pdfsam-persistence -am '-Dtest=!ZhenyuWhiteBoxTest,

# Run individual test files
mvn clean test jacoco:report -pl pdfsam-model -am '-Dtest=!ZhenyuWhiteBoxTest' '-Dsurefire.failIfNoSpecifiedT
mvn clean test jacoco:report -pl pdfsam-core -am '-Dtest=!KingsonWhiteBoxTest' '-Dsurefire.failIfNoSpecifiedT
mvn clean test jacoco:report -pl pdfsam-persistence -am '-Dtest=!ZianWhiteBoxTest' '-Dsurefire.failIfNoSpecif

# View CSV reports
cat pdfsam-model/target/site/jacoco/jacoco.csv
cat pdfsam-core/target/site/jacoco/jacoco.csv
cat pdfsam-persistence/target/site/jacoco/jacoco.csv

# View HTML reports
open pdfsam-model/target/site/jacoco/index.html
open pdfsam-core/target/site/jacoco/index.html
open pdfsam-persistence/target/site/jacoco/index.html

# Create backup directory
mkdir -p saved-reports/baseline/model saved-reports/baseline/core saved-reports/baseline/persistence

# Copy the report to the backup directory
cp -r pdfsam-model/target/site/jacoco saved-reports/baseline/model
cp -r pdfsam-core/target/site/jacoco saved-reports/baseline/core
cp -r pdfsam-persistence/target/site/jacoco saved-reports/baseline/persistence

# View CSV reports
cat saved-reports/baseline/model/jacoco/jacoco.csv
cat saved-reports/baseline/core/jacoco/jacoco.csv
cat saved-reports/baseline/persistence/jacoco/jacoco.csv

# View HTML reports
open saved-reports/baseline/model/jacoco/index.html
open saved-reports/baseline/core/jacoco/index.html
open saved-reports/baseline/persistence/jacoco/index.html
```

2.2 Modules Under Test

Module	Description	Existing Tests
`pdfsam-model`	Domain model classes	✅
`pdfsam-core`	Core utilities	✅
`pdfsam-persistence`	Data persistence layer	✅

2.3 Baseline Coverage Data

pdfsam-model

Class	Lines Missed	Lines Covered	Line Coverage	Branches Missed	Branches Covered	Branch Coverage
PdfDocumentDescriptor	13	36	73%	3	3	50%
PdfDescriptorLoadingStatus	0	33	100%	0	2	100%
ToolDescriptorBuilder	4	15	79%	0	0	—
ToolDescriptor	1	14	93%	2	0	0%
PdfRotationInput	8	13	62%	4	2	33%
BulkRotateParameters	6	13	68%	0	4	100%
Workspace	1	12	92%	0	8	100%
ToolCategory	2	10	83%	0	0	—
ToolPriority	0	8	100%	0	0	—
ObservableAtomicReference	0	8	100%	0	0	—
BaseToolBound	1	4	80%	0	0	—
SetTitleRequest	3	0	0%	0	0	—
NonExistingOutputDirectoryEvent	3	0	0%	0	0	—
NewsData	3	0	0%	0	0	—
ComboItem	15	0	0%	4	0	0%
FileType	14	0	0%	0	0	—
LatestNewsResponse	2	1	33%	0	0	—

To extract line coverage from the CSV report:

```
$ awk -F, 'NR>1 {printf "%-45s line_miss=%s line_cov=%s br_miss=%s br_cov=%s\n", $3, $8, $9, $6, $7}' saved-r
```

```
ComboItem                                     line_miss=15 line_cov=0 br_miss=4 br_cov=0
FileType                                      line_miss=14 line_cov=0 br_miss=0 br_cov=0
PdfDocumentDescriptor                         line_miss=13 line_cov=36 br_miss=3 br_cov=3
PdfRotationInput                              line_miss=8 line_cov=13 br_miss=4 br_cov=2
BulkRotateParameters                          line_miss=6 line_cov=13 br_miss=0 br_cov=4
DefaultPdfVersionComboItem                    line_miss=5 line_cov=0 br_miss=2 br_cov=0
ToolDescriptorBuilder                         line_miss=4 line_cov=15 br_miss=0 br_cov=0
ToolIdNamePair                                line_miss=3 line_cov=0 br_miss=0 br_cov=0
StageMode                                     line_miss=3 line_cov=3 br_miss=2 br_cov=0
SetTitleRequest                               line_miss=3 line_cov=0 br_miss=0 br_cov=0
RequiredPdfData                               line_miss=3 line_cov=0 br_miss=0 br_cov=0
OpenType                                      line_miss=3 line_cov=0 br_miss=0 br_cov=0
NonExistingOutputDirectoryEvent               line_miss=3 line_cov=0 br_miss=0 br_cov=0
NewsData                                      line_miss=3 line_cov=0 br_miss=0 br_cov=0
FilesDroppedEvent                             line_miss=3 line_cov=0 br_miss=0 br_cov=0
ClearToolRequest                              line_miss=3 line_cov=0 br_miss=0 br_cov=0
WorkspaceLoadedEvent                          line_miss=2 line_cov=1 br_miss=0 br_cov=0
ToolCategory                                  line_miss=2 line_cov=10 br_miss=0 br_cov=0
ShowPdfDescriptorRequest                      line_miss=2 line_cov=1 br_miss=0 br_cov=0
SetLatestStageStatusRequest                   line_miss=2 line_cov=1 br_miss=0 br_cov=0
SetActiveContentItemRequest                   line_miss=2 line_cov=1 br_miss=0 br_cov=0
RemovePdfVersionConstraintEvent               line_miss=2 line_cov=1 br_miss=0 br_cov=0
PremiumToolsResponse                          line_miss=2 line_cov=1 br_miss=0 br_cov=0
PremiumProduct                                line_miss=2 line_cov=8 br_miss=0 br_cov=0
NewImportantNewsEvent                         line_miss=2 line_cov=1 br_miss=0 br_cov=0
LoadWorkspaceRequest                          line_miss=2 line_cov=1 br_miss=0 br_cov=0
LatestNewsResponse                            line_miss=2 line_cov=1 br_miss=0 br_cov=0
AddPdfVersionConstraintEvent                  line_miss=2 line_cov=1 br_miss=0 br_cov=0
WorkspaceCloseEvent                           line_miss=1 line_cov=0 br_miss=0 br_cov=0
Workspace                                     line_miss=1 line_cov=12 br_miss=0 br_cov=8
UpdateCheckRequest                            line_miss=1 line_cov=0 br_miss=0 br_cov=0
ToolDescriptor                                line_miss=1 line_cov=14 br_miss=2 br_cov=0
Tool                                          line_miss=1 line_cov=0 br_miss=0 br_cov=0
ToggleNewsPanelRequest                        line_miss=1 line_cov=0 br_miss=0 br_cov=0
StartupEvent                                  line_miss=1 line_cov=0 br_miss=0 br_cov=0
ShutdownEvent                                 line_miss=1 line_cov=0 br_miss=0 br_cov=0
ShowStageRequest                              line_miss=1 line_cov=0 br_miss=0 br_cov=0
ShowLogMessagesRequest                        line_miss=1 line_cov=0 br_miss=0 br_cov=0
SaveLogRequest                                line_miss=1 line_cov=0 br_miss=0 br_cov=0
NoUpdateAvailable                             line_miss=1 line_cov=0 br_miss=0 br_cov=0
HideStageRequest                              line_miss=1 line_cov=0 br_miss=0 br_cov=0
HideNewsPanelRequest                          line_miss=1 line_cov=0 br_miss=0 br_cov=0
FetchPremiumModulesRequest                    line_miss=1 line_cov=0 br_miss=0 br_cov=0
FetchLatestNewsRequest                        line_miss=1 line_cov=0 br_miss=0 br_cov=0
ContentItem                                   line_miss=1 line_cov=0 br_miss=0 br_cov=0
ConfirmSaveWorkspaceRequest                   line_miss=1 line_cov=0 br_miss=0 br_cov=0
ClearLogRequest                               line_miss=1 line_cov=0 br_miss=0 br_cov=0
```

```
CleanupRequest                                  line_miss=1 line_cov=0 br_miss=0 br_cov=0
ChangedSelectedPdfVersionEvent                  line_miss=1 line_cov=0 br_miss=0 br_cov=0
BaseToolBound                                   line_miss=1 line_cov=4 br_miss=0 br_cov=0
UpdateAvailableEvent                            line_miss=0 line_cov=3 br_miss=0 br_cov=0
ToolPriority                                    line_miss=0 line_cov=8 br_miss=0 br_cov=0
ToolInputOutputType                             line_miss=0 line_cov=4 br_miss=0 br_cov=0
TaskExecutionRequest                            line_miss=0 line_cov=4 br_miss=0 br_cov=0
StageStatus                                     line_miss=0 line_cov=6 br_miss=0 br_cov=0
StageMode.new StageMode() {...}                 line_miss=0 line_cov=3 br_miss=0 br_cov=0
StageMode.new StageMode() {...}                 line_miss=0 line_cov=2 br_miss=0 br_cov=0
SetDestinationRequest                           line_miss=0 line_cov=5 br_miss=0 br_cov=0
SaveWorkspaceRequest                            line_miss=0 line_cov=8 br_miss=0 br_cov=0
PremiumTool                                     line_miss=0 line_cov=6 br_miss=0 br_cov=0
PdfLoadRequest                                  line_miss=0 line_cov=5 br_miss=0 br_cov=0
PdfFilesListLoadRequest                         line_miss=0 line_cov=4 br_miss=0 br_cov=0
PdfDescriptorLoadingStatus                      line_miss=0 line_cov=33 br_miss=0 br_cov=2
ObservableAtomicReference                       line_miss=0 line_cov=8 br_miss=0 br_cov=0
NativeOpenUrlRequest                            line_miss=0 line_cov=3 br_miss=0 br_cov=0
NativeOpenFileRequest                           line_miss=0 line_cov=3 br_miss=0 br_cov=0
LoadWorkspaceResponse                           line_miss=0 line_cov=6 br_miss=0 br_cov=0
InputPdfArgumentsLoadRequest                    line_miss=0 line_cov=2 br_miss=0 br_cov=2
```

pdfsam-core

Class	Lines Missed	Lines Covered	Line Coverage	Branches Missed	Branches Covered	Branch Coverage
ApplicationRuntimeState	17	27	61%	2	4	67%
ConversionUtils	8	29	78%	3	13	81%

To extract line coverage from the CSV report:

```
$ awk -F, 'NR>1 {printf "%-45s line_miss=%s line_cov=%s br_miss=%s br_cov=%s\n", $3, $8, $9, $6, $7}' saved-r

FileChooserWithWorkingDirectory               line_miss=35 line_cov=0 br_miss=10 br_cov=0
BrandableProperty                             line_miss=29 line_cov=0 br_miss=0 br_cov=0
ApplicationContext                            line_miss=29 line_cov=28 br_miss=5 br_cov=1
ApplicationRuntimeState                       line_miss=17 line_cov=27 br_miss=2 br_cov=4
DirectoryChooserWithWorkingDirectory          line_miss=15 line_cov=0 br_miss=4 br_cov=0
MultiplePdfSourceMultipleOutputParametersBuilder line_miss=12 line_cov=0 br_miss=2 br_cov=0
AbstractPdfOutputParametersBuilder            line_miss=11 line_cov=0 br_miss=0 br_cov=0
SinglePdfSourceMultipleOutputParametersBuilder line_miss=10 line_cov=0 br_miss=0 br_cov=0
ConversionUtils                               line_miss=8 line_cov=29 br_miss=3 br_cov=13
SplitParametersBuilder                        line_miss=7 line_cov=0 br_miss=2 br_cov=0
Choosers                                      line_miss=6 line_cov=0 br_miss=0 br_cov=0
AbstractParametersBuilder                     line_miss=5 line_cov=0 br_miss=0 br_cov=0
ObjectCollectionWriter                        line_miss=4 line_cov=21 br_miss=0 br_cov=4
EncryptionUtils                               line_miss=4 line_cov=15 br_miss=0 br_cov=4
ApplicationPersistentSettings                 line_miss=2 line_cov=71 br_miss=3 br_cov=7
Choosers.FileChooserHolder                    line_miss=1 line_cov=0 br_miss=0 br_cov=0
Choosers.DirectoryChooserHolder               line_miss=1 line_cov=0 br_miss=0 br_cov=0
XmlUtils                                      line_miss=0 line_cov=5 br_miss=0 br_cov=4
Validators                                    line_miss=0 line_cov=16 br_miss=0 br_cov=8
StringPersistentProperty                      line_miss=0 line_cov=20 br_miss=0 br_cov=0
RegexValidator                                line_miss=0 line_cov=5 br_miss=0 br_cov=4
PositiveIntRangeStringValidator               line_miss=0 line_cov=11 br_miss=2 br_cov=8
PositiveIntegerValidator                      line_miss=0 line_cov=2 br_miss=0 br_cov=4
PositiveIntegerStringValidator                line_miss=0 line_cov=4 br_miss=0 br_cov=2
PersistentPropertyChange                      line_miss=0 line_cov=1 br_miss=0 br_cov=0
IntegerPersistentProperty                     line_miss=0 line_cov=7 br_miss=0 br_cov=0
FileValidator                                 line_miss=0 line_cov=2 br_miss=0 br_cov=4
FileTypeValidator                             line_miss=0 line_cov=7 br_miss=1 br_cov=5
ContainedIntegerValidator                     line_miss=0 line_cov=7 br_miss=0 br_cov=0
BooleanPersistentProperty                     line_miss=0 line_cov=22 br_miss=0 br_cov=0
```

pdfsam-persistence

Class	Lines Missed	Lines Covered	Line Coverage	Branches Missed	Branches Covered	Branch Coverage
PreferencesRepository	16	49	75%	0	2	100%
DefaultEntityRepository	19	23	55%	0	4	100%
PersistenceException	4	2	33%	0	0	—
Repository	2	2	50%	0	0	—

To extract line coverage from the CSV report:

```
$ awk -F, 'NR>1 {printf "%-45s line_miss=%s line_cov=%s br_miss=%s br_cov=%s\n", $3, $8, $9, $6, $7}' saved-r
```

```
DefaultEntityRepository                       line_miss=19 line_cov=23 br_miss=0 br_cov=4
PreferencesRepository                         line_miss=16 line_cov=49 br_miss=0 br_cov=2
PersistenceException                          line_miss=4 line_cov=2 br_miss=0 br_cov=0
Repository                                    line_miss=2 line_cov=2 br_miss=0 br_cov=0
```

✨ 3. Zhenyu's White Box Testing: pdfsam-model

Test File: pdfsam-model/src/test/java/org/pdfsam/model/ZhenyuWhiteBoxTest.java

3.1 Target Classes

Class	Package
`PdfDocumentDescriptor`	`org.pdfsam.model.pdf`
`ToolDescriptorBuilder`	`org.pdfsam.model.tool`
`ToolDescriptor`	`org.pdfsam.model.tool`
`ToolCategory`	`org.pdfsam.model.tool`
`ObservableAtomicReference`	`org.pdfsam.model`
`BaseToolBound`	`org.pdfsam.model.tool`
`SetTitleRequest`	`org.pdfsam.model.ui`
`NonExistingOutputDirectoryEvent`	`org.pdfsam.model.ui`
`NewsData`	`org.pdfsam.model.news`
`LatestNewsResponse`	`org.pdfsam.model.news`
`ComboItem`	`org.pdfsam.model.ui`
`ClearToolRequest`	`org.pdfsam.model.tool`
`FilesDroppedEvent`	`org.pdfsam.model.ui.dnd`
`OpenType`	`org.pdfsam.model.io`
`RequiredPdfData`	`org.pdfsam.model.tool`

3.2 Test Implementation Examples

44 test methods targeting 20+ classes with previously uncovered code paths.

PdfDocumentDescriptor Tests

```
@Test
@DisplayName("hasPassword returns true when password set")
void descriptorHasPassword() {
    var desc = PdfDocumentDescriptor.newDescriptor(mockFile, "secret");
    assertTrue(desc.hasPassword());
}

@Test
@DisplayName("setVersion and getVersion")
void descriptorVersion() {
    var desc = PdfDocumentDescriptor.newDescriptorNoPassword(mockFile);
    desc.setVersion(PdfVersion.VERSION_1_7);
    assertEquals(PdfVersion.VERSION_1_7, desc.getVersion());
}
```

ToolDescriptor Tests

```
@Test
@DisplayName("hasInputType returns false for missing type")
void toolDescriptorHasInputTypeFalse() {
    ToolDescriptor td = ToolDescriptorBuilder.builder()
            .name("test").description("desc")
            .category(ToolCategory.SPLIT)
            .inputTypes(ToolInputOutputType.SINGLE_PDF)
            .build();
    assertFalse(td.hasInputType(ToolInputOutputType.MULTIPLE_PDF));
}

@Test
@DisplayName("supportURL")
void toolDescriptorSupportURL() {
    ToolDescriptor td = ToolDescriptorBuilder.builder()
            .name("test").description("desc")
            .category(ToolCategory.MERGE)
            .supportURL("https://example.com")
            .build();
    assertEquals("https://example.com", td.supportUrl());
}
```

ComboItem Tests

```java
@Test
@DisplayName("equals with same reference")
void comboItemEqualsSameRef() {
    ComboItem<String> item = new ComboItem<>("key1", "desc");
    assertEquals(item, item);
}

@Test
@DisplayName("equals with different type")
void comboItemEqualsDiffType() {
    ComboItem<String> item = new ComboItem<>("key1", "desc");
    assertNotEquals(item, "not a ComboItem");
}

@Test
@DisplayName("equals with same key")
void comboItemEqualsSameKey() {
    ComboItem<String> a = new ComboItem<>("key1", "desc1");
    ComboItem<String> b = new ComboItem<>("key1", "desc2");
    assertEquals(a, b);
}
```

Record & Event Validation Tests

```java
@Test
@DisplayName("valid construction")
void filesDroppedEventValid() {
    File f = new File("test.pdf");
    FilesDroppedEvent evt = new FilesDroppedEvent("toolId", true, List.of(f));
    assertEquals("toolId", evt.toolBinding());
    assertTrue(evt.acceptMultipleFiles());
    assertEquals(1, evt.files().size());
}
```

3.3 Coverage Improvement

Class	Before Lines	After Lines	Δ Lines	Before Branches	After Branches	Δ Branches
PdfDocumentDescriptor	36	44	**+8**	3	3	—
ToolDescriptorBuilder	15	19	**+4**	0	0	—
ToolDescriptor	14	15	**+1**	0	1	**+1**
ToolCategory	10	12	**+2**	0	0	—

Class	Before Lines	After Lines	Δ Lines	Before Branches	After Branches	Δ Branches
BaseToolBound	4	5	**+1**	0	0	—
SetTitleRequest	0	3	**+3**	0	0	—
NonExistingOutputDirectoryEvent	0	3	**+3**	0	0	—
NewsData	0	3	**+3**	0	0	—
ComboItem	0	15	**+15**	0	4	**+4**
LatestNewsResponse	1	3	**+2**	0	0	—
ClearToolRequest	0	3	**+3**	0	0	—
FilesDroppedEvent	0	3	**+3**	0	0	—
OpenType	0	3	**+3**	0	0	—
RequiredPdfData	0	3	**+3**	0	0	—
ChangedSelectedPdfVersionEvent	0	1	**+1**	0	0	—
UpdateCheckRequest	0	1	**+1**	0	0	—
Total			**+56**			**+5**

```
$ awk -F, 'NR>1 {printf "%-45s line_miss=%s line_cov=%s br_miss=%s br_cov=%s\n", $3, $8, $9, $6, $7}' pdfsam-r

FileType                                      line_miss=14 line_cov=0 br_miss=0 br_cov=0
PdfRotationInput                              line_miss=8 line_cov=13 br_miss=4 br_cov=2
BulkRotateParameters                          line_miss=6 line_cov=13 br_miss=0 br_cov=4
PdfDocumentDescriptor                         line_miss=5 line_cov=44 br_miss=3 br_cov=3
DefaultPdfVersionComboItem                    line_miss=5 line_cov=0 br_miss=2 br_cov=0
ToolIdNamePair                                line_miss=3 line_cov=0 br_miss=0 br_cov=0
StageMode                                     line_miss=3 line_cov=3 br_miss=2 br_cov=0
WorkspaceLoadedEvent                          line_miss=2 line_cov=1 br_miss=0 br_cov=0
ShowPdfDescriptorRequest                      line_miss=2 line_cov=1 br_miss=0 br_cov=0
SetLatestStageStatusRequest                   line_miss=2 line_cov=1 br_miss=0 br_cov=0
SetActiveContentItemRequest                   line_miss=2 line_cov=1 br_miss=0 br_cov=0
RemovePdfVersionConstraintEvent               line_miss=2 line_cov=1 br_miss=0 br_cov=0
PremiumToolsResponse                          line_miss=2 line_cov=1 br_miss=0 br_cov=0
PremiumProduct                                line_miss=2 line_cov=8 br_miss=0 br_cov=0
NewImportantNewsEvent                         line_miss=2 line_cov=1 br_miss=0 br_cov=0
LoadWorkspaceRequest                          line_miss=2 line_cov=1 br_miss=0 br_cov=0
AddPdfVersionConstraintEvent                  line_miss=2 line_cov=1 br_miss=0 br_cov=0
Workspace                                     line_miss=1 line_cov=12 br_miss=0 br_cov=8
Tool                                          line_miss=1 line_cov=0 br_miss=0 br_cov=0
FetchPremiumModulesRequest                    line_miss=1 line_cov=0 br_miss=0 br_cov=0
ContentItem                                   line_miss=1 line_cov=0 br_miss=0 br_cov=0
WorkspaceCloseEvent                           line_miss=0 line_cov=1 br_miss=0 br_cov=0
UpdateCheckRequest                            line_miss=0 line_cov=1 br_miss=0 br_cov=0
UpdateAvailableEvent                          line_miss=0 line_cov=3 br_miss=0 br_cov=0
ToolPriority                                  line_miss=0 line_cov=8 br_miss=0 br_cov=0
ToolInputOutputType                           line_miss=0 line_cov=4 br_miss=0 br_cov=0
ToolDescriptorBuilder                         line_miss=0 line_cov=19 br_miss=0 br_cov=0
ToolDescriptor                                line_miss=0 line_cov=15 br_miss=1 br_cov=1
ToolCategory                                  line_miss=0 line_cov=12 br_miss=0 br_cov=0
ToggleNewsPanelRequest                        line_miss=0 line_cov=1 br_miss=0 br_cov=0
TaskExecutionRequest                          line_miss=0 line_cov=4 br_miss=0 br_cov=0
StartupEvent                                  line_miss=0 line_cov=1 br_miss=0 br_cov=0
StageStatus                                   line_miss=0 line_cov=6 br_miss=0 br_cov=0
StageMode.new StageMode() {...}               line_miss=0 line_cov=3 br_miss=0 br_cov=0
StageMode.new StageMode() {...}               line_miss=0 line_cov=2 br_miss=0 br_cov=0
ShutdownEvent                                 line_miss=0 line_cov=1 br_miss=0 br_cov=0
ShowStageRequest                              line_miss=0 line_cov=1 br_miss=0 br_cov=0
ShowLogMessagesRequest                        line_miss=0 line_cov=1 br_miss=0 br_cov=0
SetTitleRequest                               line_miss=0 line_cov=3 br_miss=0 br_cov=0
SetDestinationRequest                         line_miss=0 line_cov=5 br_miss=0 br_cov=0
SaveWorkspaceRequest                          line_miss=0 line_cov=8 br_miss=0 br_cov=0
SaveLogRequest                                line_miss=0 line_cov=1 br_miss=0 br_cov=0
RequiredPdfData                               line_miss=0 line_cov=3 br_miss=0 br_cov=0
PremiumTool                                   line_miss=0 line_cov=6 br_miss=0 br_cov=0
PdfLoadRequest                                line_miss=0 line_cov=5 br_miss=0 br_cov=0
PdfFilesListLoadRequest                       line_miss=0 line_cov=4 br_miss=0 br_cov=0
PdfDescriptorLoadingStatus                    line_miss=0 line_cov=33 br_miss=0 br_cov=2
```

```
OpenType                                    line_miss=0 line_cov=3 br_miss=0 br_cov=0
ObservableAtomicReference                   line_miss=0 line_cov=8 br_miss=0 br_cov=0
NoUpdateAvailable                           line_miss=0 line_cov=1 br_miss=0 br_cov=0
NonExistingOutputDirectoryEvent             line_miss=0 line_cov=3 br_miss=0 br_cov=0
NewsData                                    line_miss=0 line_cov=3 br_miss=0 br_cov=0
NativeOpenUrlRequest                        line_miss=0 line_cov=3 br_miss=0 br_cov=0
NativeOpenFileRequest                       line_miss=0 line_cov=3 br_miss=0 br_cov=0
LoadWorkspaceResponse                       line_miss=0 line_cov=6 br_miss=0 br_cov=0
LatestNewsResponse                          line_miss=0 line_cov=3 br_miss=0 br_cov=0
InputPdfArgumentsLoadRequest                line_miss=0 line_cov=2 br_miss=0 br_cov=2
HideStageRequest                            line_miss=0 line_cov=1 br_miss=0 br_cov=0
HideNewsPanelRequest                        line_miss=0 line_cov=1 br_miss=0 br_cov=0
FilesDroppedEvent                           line_miss=0 line_cov=3 br_miss=0 br_cov=0
FetchLatestNewsRequest                      line_miss=0 line_cov=1 br_miss=0 br_cov=0
ConfirmSaveWorkspaceRequest                 line_miss=0 line_cov=1 br_miss=0 br_cov=0
ComboItem                                   line_miss=0 line_cov=15 br_miss=0 br_cov=4
ClearToolRequest                            line_miss=0 line_cov=3 br_miss=0 br_cov=0
ClearLogRequest                             line_miss=0 line_cov=1 br_miss=0 br_cov=0
CleanupRequest                              line_miss=0 line_cov=1 br_miss=0 br_cov=0
ChangedSelectedPdfVersionEvent              line_miss=0 line_cov=1 br_miss=0 br_cov=0
BaseToolBound                               line_miss=0 line_cov=5 br_miss=0 br_cov=0
```

✨ 4. Kingson's White Box Testing: pdfsam-core

Test File: pdfsam-core/src/test/java/org/pdfsam/core/context/KingsonWhiteBoxTest.java

4.1 Target Classes

Class	Package	Key Uncovered Code
`ConversionUtils`	`org.pdfsam.core.support.params`	Blank/null input, "last" keyword, open-ended ranges, prefixed dash
`ApplicationRuntimeState`	`org.pdfsam.core.context`	`maybeWorkingPath` null/blank/file/dir/string, `activeToolValue` , `workspace` , `theme`
`ObjectCollectionWriter`	`org.pdfsam.core.support.io`	`writeContent().to(Path)` with content and empty list

4.2 Test Implementation

22 test methods targeting boundary cases and uncovered branches.

ConversionUtils Edge Cases

```java
@Test
void toPageRangeSetBlankReturnsEmpty() {
    var result = ConversionUtils.toPageRangeSet("");
    assertTrue(result.isEmpty());
}

@Test
void toPagesSelectionSetLastPage() {
    Set<PagesSelection> result = ConversionUtils.toPagesSelectionSet("last");
    assertEquals(1, result.size());
    assertTrue(result.contains(PagesSelection.LAST_PAGE));
}

@Test
void toPageRangeSetOpenEndedRange() {
    Set<PageRange> result = ConversionUtils.toPageRangeSet("5-");
    assertEquals(1, result.size());
    PageRange range = result.iterator().next();
    assertEquals(5, range.getStart());
    assertTrue(range.isUnbounded());
}
```

ApplicationRuntimeState Tests

```java
@Test
void workingPathInitiallyEmpty() {
    ApplicationRuntimeState state = new ApplicationRuntimeState();
    assertEquals(Optional.empty(), state.workingPathValue());
}

@Test
void maybeWorkingPathWithValidDirectory(@TempDir Path tempDir) {
    ApplicationRuntimeState state = new ApplicationRuntimeState();
    state.maybeWorkingPath(tempDir);
    assertEquals(Optional.of(tempDir), state.workingPathValue());
}

@Test
void maybeWorkingPathWithRegularFile(@TempDir Path tempDir) throws IOException {
    Path file = Files.createTempFile(tempDir, "test", ".txt");
    ApplicationRuntimeState state = new ApplicationRuntimeState();
    state.maybeWorkingPath(file);
    // Should resolve to parent directory
    assertEquals(Optional.of(tempDir), state.workingPathValue());
}
```

4.3 Coverage Improvement

Class	Before Lines	After Lines	Δ Lines	Before Branches	After Branches	Δ Branches
ApplicationRuntimeState	27	33	**+6**	4	4	—
ConversionUtils	29	32	**+3**	13	16	**+3**
Total			**+9**			**+3**

```
$ awk -F, 'NR>1 {printf "%-45s line_miss=%s line_cov=%s br_miss=%s br_cov=%s\n", $3, $8, $9, $6, $7}' pdfsam-
```

```
FileChooserWithWorkingDirectory               line_miss=35 line_cov=0 br_miss=10 br_cov=0
BrandableProperty                             line_miss=29 line_cov=0 br_miss=0 br_cov=0
ApplicationContext                            line_miss=29 line_cov=28 br_miss=5 br_cov=1
DirectoryChooserWithWorkingDirectory          line_miss=15 line_cov=0 br_miss=4 br_cov=0
MultiplePdfSourceMultipleOutputParametersBuilder line_miss=12 line_cov=0 br_miss=2 br_cov=0
ApplicationRuntimeState                       line_miss=11 line_cov=33 br_miss=2 br_cov=4
AbstractPdfOutputParametersBuilder            line_miss=11 line_cov=0 br_miss=0 br_cov=0
SinglePdfSourceMultipleOutputParametersBuilder line_miss=10 line_cov=0 br_miss=0 br_cov=0
SplitParametersBuilder                        line_miss=7 line_cov=0 br_miss=2 br_cov=0
Choosers                                      line_miss=6 line_cov=0 br_miss=0 br_cov=0
ConversionUtils                               line_miss=5 line_cov=32 br_miss=0 br_cov=16
AbstractParametersBuilder                     line_miss=5 line_cov=0 br_miss=0 br_cov=0
ObjectCollectionWriter                        line_miss=4 line_cov=21 br_miss=0 br_cov=4
EncryptionUtils                               line_miss=4 line_cov=15 br_miss=0 br_cov=4
ApplicationPersistentSettings                 line_miss=2 line_cov=71 br_miss=3 br_cov=7
Choosers.FileChooserHolder                    line_miss=1 line_cov=0 br_miss=0 br_cov=0
Choosers.DirectoryChooserHolder               line_miss=1 line_cov=0 br_miss=0 br_cov=0
XmlUtils                                      line_miss=0 line_cov=5 br_miss=0 br_cov=4
Validators                                    line_miss=0 line_cov=16 br_miss=0 br_cov=8
StringPersistentProperty                      line_miss=0 line_cov=20 br_miss=0 br_cov=0
RegexValidator                                line_miss=0 line_cov=5 br_miss=0 br_cov=4
PositiveIntRangeStringValidator               line_miss=0 line_cov=11 br_miss=2 br_cov=8
PositiveIntegerValidator                      line_miss=0 line_cov=2 br_miss=0 br_cov=4
PositiveIntegerStringValidator                line_miss=0 line_cov=4 br_miss=0 br_cov=2
PersistentPropertyChange                      line_miss=0 line_cov=1 br_miss=0 br_cov=0
IntegerPersistentProperty                     line_miss=0 line_cov=7 br_miss=0 br_cov=0
FileValidator                                 line_miss=0 line_cov=2 br_miss=0 br_cov=4
FileTypeValidator                             line_miss=0 line_cov=7 br_miss=1 br_cov=5
ContainedIntegerValidator                     line_miss=0 line_cov=7 br_miss=0 br_cov=0
BooleanPersistentProperty                     line_miss=0 line_cov=22 br_miss=0 br_cov=0
```

✨ 5. Zian's White Box Testing: pdfsam-persistence

Test File: pdfsam-persistence/src/test/java/org/pdfsam/persistence/ZianWhiteBoxTest.java

5.1 Target Classes

Class	Package	Key Uncovered Code
`PersistenceException`	`org.pdfsam.persistence`	Two constructors: message-only, cause-only
`DefaultEntityRepository`	`org.pdfsam.persistence`	Primitive type accessors: `saveInt` / `getInt` , `saveLong` / `getLong` , `saveString` / `getString` , `saveBoolean` / `getBoolean` , `keys()`
`PreferencesRepository`	`org.pdfsam.persistence`	`saveInt` / `getInt` , `saveLong` / `getLong` , `saveBoolean` / `getBoolean` , `keys()` , `clean`
`Repository`	`org.pdfsam.persistence`	`getInt` , `getLong`

5.2 Test Implementation

27 test methods were implemented to achieve comprehensive coverage, specifically targeting primitive types, interface default methods, and hard-to-reach exception handling blocks.

PersistenceException Tests

```java
@Test
@DisplayName("PersistenceException: message-only constructor")
void persistenceExceptionMessageOnly() {
    PersistenceException ex = new PersistenceException("test error");
    assertEquals("test error", ex.getMessage());
    assertNull(ex.getCause());
}
```

```java
@Test
@DisplayName("PersistenceException: cause-only constructor")
void persistenceExceptionCauseOnly() {
    Throwable cause = new RuntimeException("root cause");
    PersistenceException ex = new PersistenceException(cause);
    assertEquals(cause, ex.getCause());
}
```

DefaultEntityRepository Primitive Accessors

```java
@Test
@DisplayName("DefaultEntityRepository: saveInt and getInt")
void entityRepoSaveGetInt() throws PersistenceException {
    entityRepo.saveInt("intKey", 42);
    assertEquals(42, entityRepo.getInt("intKey", 0));
}

@Test
@DisplayName("DefaultEntityRepository: getInt default value")
void entityRepoGetIntDefault() throws PersistenceException {
    entityRepo.delete("intMissing");
    assertEquals(-1, entityRepo.getInt("intMissing", -1));
}

@Test
@DisplayName("DefaultEntityRepository: saveLong and getLong")
void entityRepoSaveGetLong() throws PersistenceException {
    entityRepo.saveLong("longKey", 123456789L);
    assertEquals(123456789L, entityRepo.getLong("longKey", 0L));
}
```

Repository Tests

```java
@Test
@DisplayName("Repository: getInt uses Supplier for default value")
void repositoryGetIntWithSupplier() throws PersistenceException {
    entityRepo.delete("intSuppMissing");

    assertEquals(100, entityRepo.getInt("intSuppMissing", () -> 100));
}

@Test
@DisplayName("Repository: getLong uses Supplier for default value")
void repositoryGetLongWithSupplier() throws PersistenceException {
    entityRepo.delete("longSuppMissing");

    assertEquals(999L, entityRepo.getLong("longSuppMissing", () -> 999L));
}
```

PreferencesRepository Tests

```java
@Nested
class ZianExceptionCoverageTest {
    private final PreferencesRepository brokenRepo = new PreferencesRepository("/invalid//double/slash/path")

    @Test
    @DisplayName("getInt triggers PersistenceException on invalid path")
    void getIntException() {
        assertThrows(PersistenceException.class, () -> brokenRepo.getInt("anyKey", 0));
    }

    @Test
    @DisplayName("getLong triggers PersistenceException on invalid path")
    void getLongException() {
        assertThrows(PersistenceException.class, () -> brokenRepo.getLong("anyKey", 0L));
    }
}
```

5.3 Coverage Improvement

Class	Before Lines	After Lines	Δ Lines	Before Branches	After Branches	Δ Branches
DefaultEntityRepository	23	42	**+19**	4	4	—
PreferencesRepository	49	61	**+12**	2	2	—
PersistenceException	2	6	**+4**	0	0	—
Repository	2	4	**+2**	0	0	—
Total			**+37**			**0**

```
$ awk -F, 'NR>1 {printf "%-45s line_miss=%s line_cov=%s br_miss=%s br_cov=%s\n", $3, $8, $9, $6, $7}' pdfsam-
```

```
PreferencesRepository                         line_miss=4 line_cov=61 br_miss=0 br_cov=2
Repository                                    line_miss=0 line_cov=4 br_miss=0 br_cov=0
PersistenceException                          line_miss=0 line_cov=6 br_miss=0 br_cov=0
DefaultEntityRepository                       line_miss=0 line_cov=42 br_miss=0 br_cov=4
```

📋 6. Test Implementation Summary

6.1 New Test Files

File	Location	Author
ZhenyuWhiteBoxTest.java	`pdfsam-model/src/test/java/org/pdfsam/model/`	Zhenyu Song
KingsonWhiteBoxTest.java	`pdfsam-core/src/test/java/org/pdfsam/core/context`	Kingson Zhang
ZianWhiteBoxTest.java	`pdfsam-persistence/src/test/java/org/pdfsam/persistence/`	Zian Xu

6.2 Running the White Box Tests

```
# Run all white box tests together
mvn clean test jacoco:report -pl pdfsam-model,pdfsam-core,pdfsam-persistence -am

# Run individual test files
mvn test jacoco:report -pl pdfsam-model -Dtest=ZhenyuWhiteBoxTest
mvn test jacoco:report -pl pdfsam-core -Dtest=KingsonWhiteBoxTest
mvn test jacoco:report -pl pdfsam-persistence -Dtest=ZianWhiteBoxTest

# View CSV reports
cat pdfsam-model/target/site/jacoco/jacoco.csv
cat pdfsam-core/target/site/jacoco/jacoco.csv
cat pdfsam-persistence/target/site/jacoco/jacoco.csv

# View HTML reports
open pdfsam-model/target/site/jacoco/index.html
open pdfsam-core/target/site/jacoco/index.html
open pdfsam-persistence/target/site/jacoco/index.html
```

6.3 Test Results

```
$ mvn test jacoco:report -pl pdfsam-model -Dtest=ZhenyuWhiteBoxTest
Tests run: 44, Failures: 0, Errors: 0, Skipped: 0
BUILD SUCCESS


$ mvn test jacoco:report -pl pdfsam-core -Dtest=KingsonWhiteBoxTest
Tests run: 28, Failures: 0, Errors: 0, Skipped: 0
BUILD SUCCESS
```

```
$ mvn test jacoco:report -pl pdfsam-persistence -Dtest=ZianWhiteBoxTest
Tests run: 27, Failures: 0, Errors: 0, Skipped: 0
BUILD SUCCESS
```

🎯 7. Conclusion

This report documents our application of **structural (white-box) testing** to PDFsam Basic through three team members testing three non-GUI modules:

Team Member	Module	Δ New Tests	Δ Lines	Δ Branches
Zhenyu Song	pdfsam-model	**+44**	**+56**	**+5**
Kingson Zhang	pdfsam-core	**+28**	**+9**	**+3**
Zian Xu	pdfsam-persistence	**+27**	**+37**	**0**
Total		**+99**	**+102**	**+8**

Key Takeaways

- **JaCoCo** provides systematic, automated coverage measurement in CSV/XML format
- **Structural testing** complements the partition testing (Part 1) and FSM testing (Part 2) by revealing untested implementation paths
- **Coverage-guided test design** helps prioritize which classes and branches need additional tests

The white-box tests complement the **partition testing** from Part 1 and **FSM testing** from Part 2, providing a third perspective on the same codebase and moving toward comprehensive test coverage.